

Assignment 01

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Analysis of Food Delivery System

a. Purpose

The system connects customers, restaurants and riders on one platform. It lets customers order and pay for food online and track delivery. It replaces slow phone ordering and gives restaurants a wider reach and riders steady work.

b. Users

- Customers: search, order, pay, track and rate
- Restaurants staff: manage menus, accept and prepare orders.
- Riders: accept deliveries and confirm drop-off.
- Admin and support: handle complaints, refunds and accounts.

c. Stakeholders

- The platform company and its investors.
- Restaurant owners
- Riders
- Customers
- Payment providers and banks.

- Governments and regulators (tax, food safety, data protection)
- Developers and maintenance teams
- Map and SMS service providers.

d. Five Functional Requirements

1. Registration and login: Users can register and login securely.
2. Search and browse: Customers can search restaurants and dishes by cuisine, price and rating.
3. Order and payment: Customers can order and pay by card, wallet or COD.
4. Order management and rider assignment: Restaurants accept orders and update their status and the system assigns a rider.
5. Live tracking and Notifications: The system shows the order status and rider location and sends updates at each stage.

F. Risks

Risks

- Data or payment breach exposes customer information and destroys trusts.
- Overload at peak times crashes the system and loses orders.
- Poor delivery or food quality (late, cold, wrong or unhygienic food) harms the reputation.

Mitigation

- Encryption, secure gateways, security testing.
- Load testing, auto-scaling, backup servers.
- Ratings, restaurants verification, tracking, fast refunds.

G. Quality Attributes

Reliability: orders must not be lost or double charged, and the system must recover quickly (backups, retries).

Usability: ordering takes only a few taps, with a simple interface that supports local languages.

Maintainability: modular, well-documented code makes updates, bug fixes and new features fast and cheap.

H. AI Could Be Used

- ♦ Recommendations: suggest dishes and restaurants from past orders.
- ♦ ETA prediction: use traffic, weather and preparation time.
- ♦ Rider assignment and route optimization: match orders to the best rider and route.
- ♦ Demand forecasting: predict busy times and (refund) areas.
- ♦ Chatbot support: answer order and refund questions 24/7.
- ♦ Fraud detection: flag fake accounts, stolen cards and refund abuse.
- ♦ Review analysis: detect fake ratings and quality problems.

I. What could go wrong if AI is used incorrectly

- ♦ Wrong prediction: inaccurate ETAs lead to angry customers.
- ♦ Bias: Big restaurants may be favoured over small ones, and riders may be treated unfairly.

- ♦ Chatbot mistakes ♦ wrong refund or allergen information can be dangerous.
- ♦ Privacy violation ♦ misuse or leaks of location, order and payment data.
- ♦ False fraud flags ♦ genuine users get blocked with no way appeal.
- ♦ Unfair pricing ♦ Hidden price differences or manipulation of users.
- ♦ Over-reliance on AI ♦ Small errors grow into big failures when no human is in charge.

Conclusion: AI can greatly improve food delivery, but it needs good data, human oversight, transparency and respect for privacy.
